

RBF Conformer for Dummies

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1 Abstract

2 Introduction

3 Mathematical Framework

3.1 Problem Setting

Given a set of subjects, each associated with a variable number of EEG trials, let $\mathcal{D} = \{(\mathbf{X}_{s,n}, y_{s,n})\}_{s=1, n=1}^{S, N_s}$ denote the dataset, where S is the total number of subjects and N_s is the number of trials for subject s . Each trial $\mathbf{X}_{s,n} \in \mathbb{R}^{C \times T}$ represents an EEG recording with C channels and T temporal samples, and $y_{s,n} \in \{0, 1\}$ denotes its corresponding label. Notably, all trials belonging to a given subject share the same label.

Given an input trial $\mathbf{X}_{s,n}$, the model outputs a probability $\hat{y}_{s,n} \in (0, 1)$ of belonging to label 1. The objective is to learn a mapping that accurately predicts the labels across all trials while accounting for the subject-wise structure of the data. In the same spirit of [1], the proposed model is structured into three modules designed to learn both local and global features across the channel and temporal dimensions.

3.2 Embedding Module-**propuesta**

Given an input trial $\mathbf{X}$, and omitting subscripts for simplicity, a bank of F filters with kernel size $(1, k_t)$ is applied in a channel-wise manner, producing a set of temporally filtered representations of the EEG trial. Subsequently, to integrate information across channels, a convolution with kernel size $(C, 1)$ is applied across all channels for each filter.

This process yields an embedding representation $\mathbf{Z} \in \mathbb{R}^{F \times T'}$, where F denotes the number of learned filters and defines the dimensionality of the latent

feature space, while T' corresponds to a reduced temporal resolution resulting from the convolutional operations. In this sense, T' can be interpreted as a downsampled version of the original temporal dimension, preserving the most relevant patterns while reducing temporal redundancy.

This formulation produces a compact representation of the EEG trial, where the original signal is projected onto a set of learned filters that capture discriminative spatio-temporal patterns. The reduced temporal resolution further lowers the computational cost of subsequent transformer layers while retaining the essential signal dynamics.

3.3 Transformer Encoder Module

From the embedding representation $\mathbf{Z}$, the model employs a transformer encoder to capture long-range dependencies between temporal segments of the EEG signal [2]. This module enables the modeling of global interactions between the embedded representations generated by the convolutional stage, overcoming the limitations of purely local methods. Each layer of the transformer encoder consists of two main components: a multi-head self-attention mechanism and a position-wise feed-forward network.

3.3.1 Self-Attention

Given the embedding representation $\mathbf{Z} \in \mathbb{R}^{F \times T'}$, the model projects it into three distinct subspaces, yielding the queries, keys, and values matrices:

$$\mathbf{Q} = \mathbf{W}_Q \mathbf{Z}, \quad \mathbf{K} = \mathbf{W}_K \mathbf{Z}, \quad \mathbf{V} = \mathbf{W}_V \mathbf{Z}, \quad (1)$$

where $\mathbf{W}_Q, \mathbf{W}_K, \mathbf{W}_V \in \mathbb{R}^{D \times F}$ are learnable projection matrices, and D corresponds to the dimensionality of each subspace. In the standard formulation, the self-attention mechanism operates on the queries, keys and values matrices via scaled dot-product operation as follow in Equation (2):

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = \text{Softmax} \left(\frac{\mathbf{Q}^\top \mathbf{K}}{\sqrt{D}} \right) \mathbf{V}^\top \in \mathbb{R}^{T' \times D}, \quad (2)$$

with $\text{Softmax}(\cdot)$ operator applied row-wise to produce normalized attention weights across the temporal dimension.

3.3.2 Multi-Head Attention

In the multi-head formulation, self-attention process described in Equation (2) is repeated across H independent attention heads, each one producing an output denoted as $\text{head}_i = \text{Attention}(\mathbf{Q}_i, \mathbf{K}_i, \mathbf{V}_i)$. Then, the outputs of all attention heads are then concatenated along D dimension:

$$\text{MHA}(\mathbf{Q}, \mathbf{K}, \mathbf{V}) = [\text{head}_1 \parallel \dots \parallel \text{head}_H] \mathbf{W}_O \in \mathbb{R}^{T' \times F}, \quad (3)$$

with $\mathbf{W}_O \in \mathbb{R}^{F \times F}$ as output projection matrix. In this way, each head capture different temporal dependencies and integrate them with a linear projection. However, the above formulation impose the constraint $D = \frac{F}{H} \in \mathbb{N}$, i.e, the number of filters F must be a multiple of the number of heads H . To overcoming this restriction, the embedding representation $\mathbf{Z}$ is first linearly projected into a suitable embedding space through a learnable transformation matrix before MHA operation.

3.3.3 Position-Wise Feed-Forward and Classification Head

Following the multi-head self-attention mechanism, the resulting representation undergoes processing by a position-wise feed-forward network applied independently to each temporal position. Additionally, residual connections are further incorporated around both the multi-head attention and feed-forward sublayers. The above constitute a single transformer encoder layer, which can be stacked to an arbitrary numbers of layers.

For the classification task, the output of the last transformer encoder layer is flattened into a one-dimensional vector and passed through a multi-layer perceptron. Finally, a sigmoid activation the produces the probability to EEG trial of belonging to class 1, i.e., $P(\hat{y} = 1 \mid \mathbf{X})$.

3.4 Kernel-Based Transformer Encoder -Proposed

Starting from the formulation given in Equation (2), we adopt a generalized similarity-based framework in which the interaction between queries and keys is defined through a kernel function, rather than being restricted to the standard dot-product [3]. The attention operator is defined in a row-wise manner, where each i -th row corresponds to the output given by Equation (4).

$$\text{Attention}(\mathbf{Q}, \mathbf{K}, \mathbf{V})_i = \sum_{j=1}^{T'} \frac{k(\mathbf{Q}_i, \mathbf{K}_j)}{\sum_{\ell=1}^{T'} k(\mathbf{Q}_i, \mathbf{K}_\ell)} \mathbf{V}_j \in \mathbb{R}^D, \quad (4)$$

where $\mathbf{Q}_i, \mathbf{K}_j, \mathbf{V}_j \in \mathbb{R}^D$ denote the i -th and j -th row vectors of the matrices $\mathbf{Q}, \mathbf{K}, \mathbf{V}$, respectively, as defined in Equation (1). The function k represents a kernel that quantifies pairwise similarity between the query $\mathbf{Q}_i$ and the key $\mathbf{K}_j$. This formulation enables the model to flexibly adapt interactions across temporal elements, thereby enhancing its capacity to capture complex and non-linear relationships.

Since the query, key, and value matrices are obtained as linear projections of the embedding $\mathbf{Z}$ according to Equation (1), each column $\mathbf{z}_j \in \mathbb{R}^F$ represents the feature vector associated with the j -th temporal position. In this sense, the attention mechanism can be reinterpreted directly in the embedding space, where interactions between representations are fully determined by similarity relationships between $\mathbf{z}_i$ and $\mathbf{z}_j$.

Under this perspective, the normalized form of the kernel in Equation (4) naturally induces a normalized weighting scheme over the samples, which can be

99 interpreted as a conditional distribution over the available representations. In
100 particular, these weights can be understood as an approximation of the condi-
101 tional density $p(\mathbf{z} \mid \mathbf{z}_i)$. Since this density is unknown, it is approximated using
102 a Parzen window estimator based on the available samples $\{\mathbf{z}_1, \dots, \mathbf{z}_{T'}\}$. Under
103 this approximation, the discrete distribution induced by the kernel is defined
104 as:

$$P(\mathbf{z}_j \mid \mathbf{z}_i) \approx \frac{k(\mathbf{z}_i, \mathbf{z}_j; \mathbf{W}_Q, \mathbf{W}_K)}{\sum_{j'=1}^{T'} k(\mathbf{z}_i, \mathbf{z}_{j'}; \mathbf{W}_Q, \mathbf{W}_K)} \quad (5)$$

105 where $k : \mathbb{R}^F \times \mathbb{R}^F \rightarrow \mathbb{R}^+$ is a positive kernel function parameterized by the
106 projection matrices $\mathbf{W}_Q$ and $\mathbf{W}_K$. Under this formulation, the attention mecha-
107 nism can be interpreted as the expected value of the transformed representations
108 $\mathbf{W}_V \mathbf{z}_j$ with respect to the kernel-induced distribution:

$$\text{Attention}(\mathbf{Z})_i = \sum_{j=1}^{T'} P(\mathbf{z}_j \mid \mathbf{z}_i) \mathbf{W}_V \mathbf{z}_j = \mathbf{W}_V \mathbb{E}_{P(\mathbf{z}_j \mid \mathbf{z}_i)}[\mathbf{z}_j] \quad (6)$$

109 In this way, kernel-based attention admits a clear probabilistic interpreta-
110 tion: the output $\text{Attention}(\mathbf{Z})_i$ corresponds to the expected value of the value
111 representations under a probability distribution induced by the kernel.

112 4 Experimental Setup

113 4.1 Dataset Description

114 The study employs multichannel electroencephalographic (EEG) recordings from
115 the public dataset “EEG Data for ADHD / Control Children” [4]. The database
116 contains signals from $S = 121$ subjects, of which 61 belong to the ADHD class
117 and 60 to the control class. Each subject is assigned a binary label $y_s \in \{0, 1\}$,
118 where 0 denotes control and 1 denotes ADHD.

119 EEG acquisition follows the international 10–20 system using 19 electrodes
120 distributed across frontal, central, temporal, parietal, and occipital regions, as
121 illustrated in Figure 1. The recorded channels include Fp1, Fp2, F3, F4, F7, F8,
122 Fz, Cz, Pz, C3, C4, T3, T4, T5, T6, P3, P4, O1, and O2, with references placed
123 on the earlobes (A1, A2). Signals were sampled at a frequency of $f_s = 128$
124 Hz. Each recording corresponds to a continuous EEG signal acquired during a
125 visual attention paradigm, in which participants perform a counting task based
126 on sequentially presented visual stimuli.

127 4.2 Preprocessing

128 The preprocessing pipeline transforms continuous EEG recordings into short-
129 duration trials suitable for model training. A Butterworth bandpass filter in
130 the range $[0.5, 60]$ Hz removes low-frequency drifts and high-frequency noise.
131 Furthermore, a notch filter at 50 Hz suppresses power-line interference.

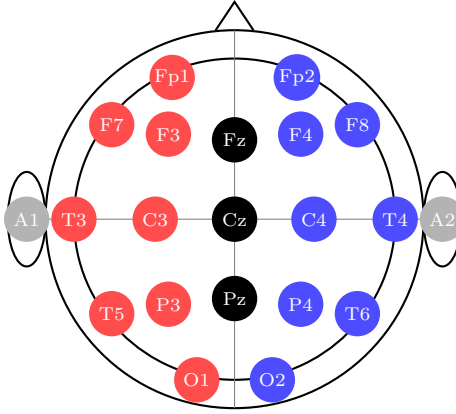


Figure 1: EEG acquisition setup under the 10–20 system.

132 The filtered signal is subsequently segmented using a sliding window of length
 133 $L = 2$ seconds with 50% overlap. This segmentation produces multiple trials
 134 per subject, consistent with the formulation described in Section 3.1. The use
 135 of overlapping windows increases the number of training samples and maintains
 136 temporal continuity between adjacent segments, improving model robustness in
 137 scenarios with limited subject availability and non-stationary EEG dynamics
 138 [5, 6].

139 Figure 2 illustrates the distribution of N_s across subjects, showing variability
 140 in the number of generated trials due to differences in recording duration.

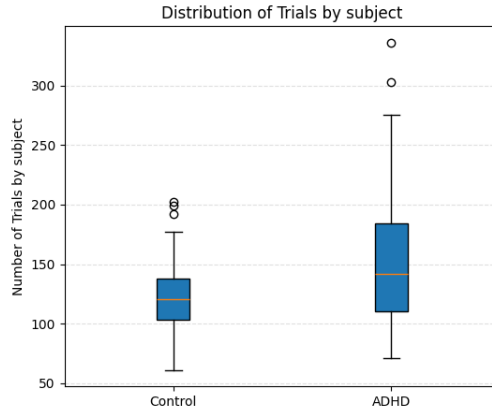


Figure 2: Distribution of trials per subject.

4.3 Nested Group K-Fold Evaluation

Model evaluation follows a nested group-based cross-validation scheme. Let $\mathcal{S}$ denote the set of subjects. A Group K -Fold partition with $K = 10$ divides $\mathcal{S}$ into disjoint subsets $\{\mathcal{S}^{(k)}\}_{k=1}^K$. At each outer iteration k , the subset $\mathcal{S}^{(k)}$ defines the test set, while the remaining subjects form the development set. A second split is then applied to the development set to separate training and validation subsets, preserving subject-level grouping.

Model parameters are optimised by minimising the binary cross-entropy loss:

$$l_{\text{BCE}}(y_{s,n}, \tilde{y}_{s,n}) = \sum_{n=1}^{N_s} [y_{s,n} \log(\tilde{y}_{s,n}) + (1 - y_{s,n}) \log(1 - \tilde{y}_{s,n})] \quad (7)$$

$$\mathcal{L} = - \sum_{s=1}^S l_{\text{BCE}}(y_{s,n}, \tilde{y}_{s,n}) \quad (8)$$

Performance metrics, including accuracy, precision, recall, and F1-score, were computed from the confusion matrix. Since each subject is associated with multiple trials and all trials inherit the subject-level label, the evaluation is conducted at the subject level. To this end, a majority voting scheme is applied to the model predictions, as shown below:

$$\hat{y}_s = \arg \max_{c \in \{0,1\}} \sum_{n=1}^{N_s} l_{\text{BCE}}(c, \tilde{y}_{s,n}) \quad (9)$$

4.4 Model Configuration

Hyperparameter selection is performed using Bayesian optimisation over the validation sets. This process explores configurations that balance model capacity and generalisation. The model consists of three main components: (i) a convolutional embedding module, (ii) a transformer encoder with a kernel-based attention mechanism, and (iii) a classification head.

The architecture incorporates non-linear activation functions, layer normalisation within transformer blocks, and dropout as a regularisation mechanism. These components support stable training and reduce overfitting without increasing model complexity. The optimisation process tunes the main hyperparameters of the model, including: Embedding dimension F' , Number of transformer layers L_{tr} , Number of attention heads H , Feed-forward dimension d_{ff} , Number of convolutional filters F , Temporal kernel size k_t , Pooling parameters (k_p, s_p) , Dropout rate p .

In particular, the attention mechanism replaces the standard softmax normalisation with a kernel-based formulation, enabling more flexible similarity measures between query and key representations. This formulation is well suited for EEG signals, which exhibit non-stationary behaviour and nonlinear dependencies. For this purpose, several kernel functions were evaluated, as shown in Table 1.

Table 1: Kernel functions

Kernel	Expression
Standard exponential	$k(\mathbf{Q}_i, \mathbf{K}_j) = \exp(\mathbf{Q}_i^\top \mathbf{K}_j)$
RBF	$k(\mathbf{Q}_i, \mathbf{K}_j) = \exp(-\ \mathbf{Q}_i - \mathbf{K}_j\ ^2)$
Periodic	$k(\mathbf{Q}_i, \mathbf{K}_j) = \exp(-2 \sin^2(\ \mathbf{Q}_i - \mathbf{K}_j\ _2))$

174 To provide a structured overview of the proposed architecture, Table 2 sum-
175 marises the main components of the model, including the embedding stage,
176 transformer encoder, and classification head, along with their corresponding
177 output dimensions and key parameters.

Table 2: Proposed EEG Conformer Architecture

Layer	Output Size	Parameters
Input	EEG Segment	(C, T) –
Patch Embedding		
Temporal Conv	(F, C, T')	$F \cdot k_t$
Spatial Conv	$(F, 1, T')$	$F^2 \cdot C$
Projection	$(F', 1, T')$	–
Transformer Encoder ($\times L$ layers)		
Multi-Head Attention	$(F_{\text{emb}}, T_{\text{emb}})$	–
Add & Norm	$(F_{\text{emb}}, T_{\text{emb}})$	–
Feed Forward	$2F_{\text{emb}}d_{ff}$	–
Add & Norm	$(F_{\text{emb}}, T_{\text{emb}})$	–
Classification Head		
Flatten	$(F \cdot T')$	–
FC1	(256)	–
FC2	(32)	–
Output	(K)	classes

178 5 Results and discussion

179 5.1 Quantitative Results

180 Table 3 reports the classification performance of the model under three kernel-
181 based attention formulations, considering the exponential kernel as the baseline.

182 The results indicate that the RBF kernel consistently achieves the best
183 performance across all evaluated metrics. Additionally, this configuration ex-
184 hibits lower variability across runs, suggesting improved stability with respect
185 to weight initialization and the optimization process. This behavior is partic-
186 ularly evident in the F1-score, where the RBF-based model shows a standard
187 deviation of ± 0.10 , compared to ± 0.17 for the exponential kernel.

188 The precision–recall trade-off further highlights distinct decision behaviors
 189 induced by each kernel. The periodic kernel attains a high recall (0.94 ± 0.11),
 190 but at the expense of a substantial drop in precision (0.67 ± 0.22), indicating a
 191 higher false positive rate and a tendency toward less restrictive decision bound-
 192 aries. In contrast, the RBF kernel achieves a more balanced trade-off, with
 193 precision and recall values of 0.79 ± 0.12 and 0.95 ± 0.08 , respectively, resulting
 194 in the highest overall F1-score.

195 Overall, these results demonstrate that the choice of similarity function
 196 within the attention mechanism has a direct impact on the model’s discrim-
 197 inative capability. In particular, the RBF kernel enables the learning of more
 198 robust and stable representations, improving both average performance and
 199 consistency across runs.

Table 3: Performance comparison across different attention kernels

Kernel	Accuracy	F1-score	Precision	Recall	AUC
Exponential	0.81 ± 0.14	0.82 ± 0.17	0.75 ± 0.17	0.91 ± 0.19	0.89 ± 0.08
Periodic	0.69 ± 0.23	0.76 ± 0.16	0.67 ± 0.22	0.94 ± 0.11	0.78 ± 0.24
RBF	0.84 ± 0.10	0.86 ± 0.10	0.79 ± 0.12	0.95 ± 0.08	0.90 ± 0.09

200 5.2 Embedding Space Analysis

201 To better understand how the proposed model transforms the feature space, a
 202 t-SNE projection is computed at different stages of the transformer, including
 203 the input embedding, intermediate layers, and the final representation. Fig-
 204 ure 3 illustrates the evolution of the embedding space across layers. At the
 205 input stage, representations from both classes (Control and ADHD) exhibit a
 206 high degree of overlap, indicating that the initial feature space is not linearly
 207 separable and lacks a clear discriminative structure.

208 As the representations propagate through the transformer layers, a progres-
 209 sive organization of the feature space can be observed. In the early layers,
 210 partial grouping begins to emerge, although a considerable degree of inter-class
 211 mixing remains. At intermediate depths, the embeddings start to form more
 212 structured manifolds, where samples belonging to the same class become in-
 213 creasingly localized, suggesting the emergence of latent patterns relevant to the
 214 classification task. In deeper layers, this effect becomes more pronounced, with
 215 class-specific clusters appearing more distinctly separated. This behavior in-
 216 dicates that the transformer encoder learns hierarchical representations that
 217 emphasize discriminative features while suppressing irrelevant variability.

218 At the output of the model, the embedding space exhibits compact and
 219 well-defined clusters, reflecting a high degree of class separability. Nevertheless,
 220 a transition region characterized by partial overlap between classes can still
 221 be observed. This region can be interpreted as an uncertainty zone, where
 222 the model exhibits lower confidence in label assignment. In contrast, samples
 223 located farther from this boundary tend to correspond to predictions with higher

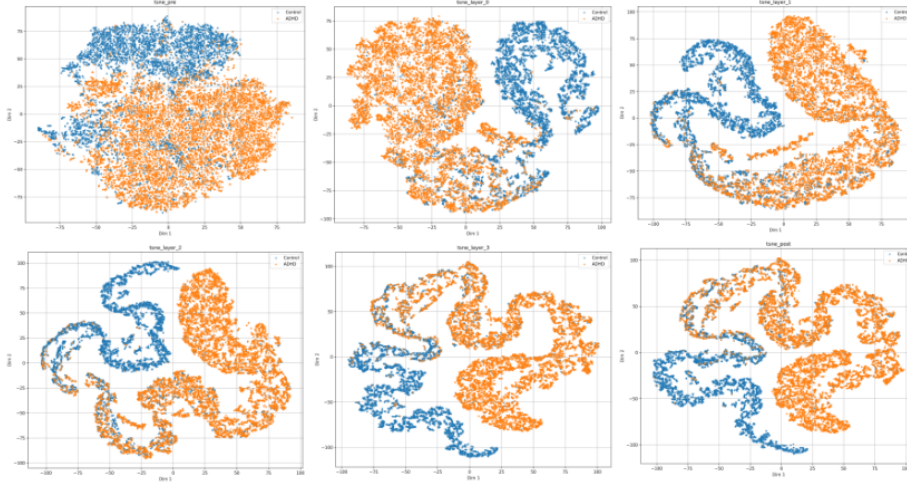


Figure 3: Caption

confidence, suggesting that the geometry of the latent space is aligned with the model’s confidence estimates.

Additionally, a higher dispersion is observed in the representations associated with the ADHD class. This behavior may be attributed to the inherently heterogeneous nature of this population, which is characterized by greater inter-subject variability in neurophysiological patterns. Furthermore, factors such as differences in symptom severity, clinical subtypes, or the presence of noise in EEG recordings may contribute to a more spread-out representation in the latent space. Overall, this behavior supports the hypothesis that the attention mechanism, particularly under the proposed kernel-based formulation, effectively reshapes the feature space into a more discriminative representation. Taken together, these results provide qualitative evidence that the model enhances class separability across layers, transforming an initially entangled representation into a structured latent space suitable for classification.

5.3 Statistical Significance Analysis **Pendiente por redactar**

To assess whether the observed performance differences are statistically significant, a permutation test is conducted between the best-performing baseline and the proposed model. +

Both models are trained multiple times using different random seeds to account for variability in the optimization process.

Figure 4 illustrates the null distribution obtained from the permutation procedure, along with the observed difference in performance.

The resulting p-value is significantly below conventional thresholds (e.g., $p < 0.05$), indicating that the improvement achieved by the proposed model is

249 unlikely to be due to random chance.

250 Additionally, the effect size (Cohen's $d = X$) suggests a moderate to strong
 251 practical significance, reinforcing the robustness of the proposed approach.

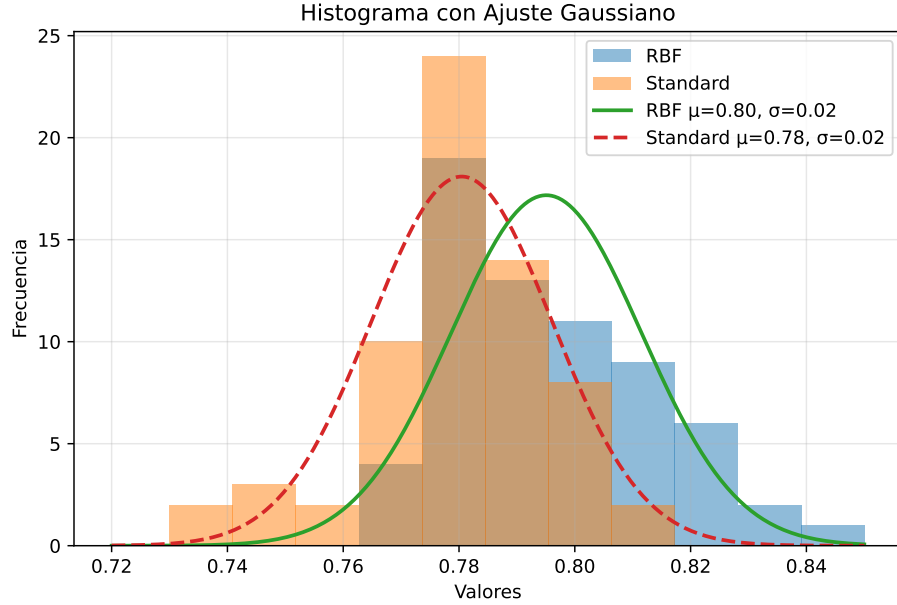


Figure 4: Caption

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 253 diente mis hyperpametros vienen de las desiciones tomadas de esa confuguracion
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255 5.4 Future work

256 References

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